1.9.5

EE25BTECH11018 - Darisy Sreetej

Question:
The distance between the points $\left(0, 2\sqrt{5}\right)$ and $\left(-2\sqrt{5}, 0\right)$ is ____________.
Solution:
The input parameters for this problem are available in Table **??**.

Symbol	Value	Description
P	$\begin{pmatrix} 0 \\ 2\sqrt{5} \end{pmatrix}$	First point
Q	$\begin{pmatrix} -2\sqrt{5} \\ 0 \end{pmatrix}$	Second point
d	?	Required distance

TABLE I: Input parameters

The distance between **P** and **Q** is given by

$$d = \|\mathbf{P} - \mathbf{Q}\| \tag{1}$$

$$\begin{aligned}\mathbf{P} - \mathbf{Q} &= \begin{pmatrix} 0 \\ 2\sqrt{5} \end{pmatrix} - \begin{pmatrix} -2\sqrt{5} \\ 0 \end{pmatrix} \\ &= \begin{pmatrix} 2\sqrt{5} \\ 2\sqrt{5} \end{pmatrix}\end{aligned}$$

Substituting in (**??**),

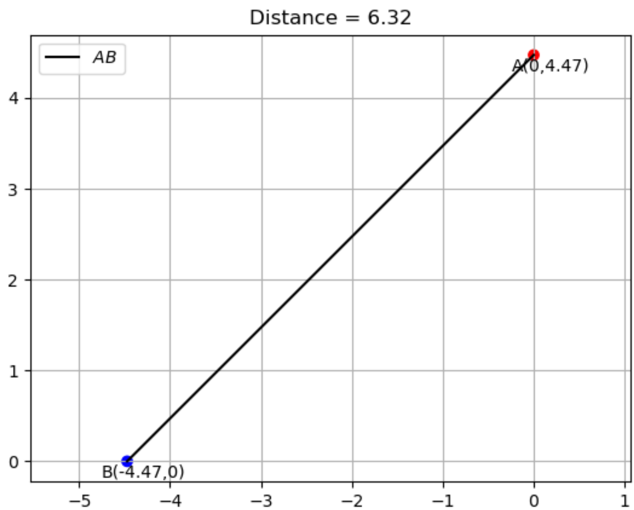
$$d = \sqrt{(2\sqrt{5})^2 + (2\sqrt{5})^2} \tag{2}$$

$$= \sqrt{20 + 20} \tag{3}$$

$$= \sqrt{40} \tag{4}$$

$$= 2\sqrt{10} \tag{5}$$

$\therefore$ The required distance is $\mathbf{2}\sqrt{10}$


Distance = 6.32
AB
A(0,4.47)
B(-4.47,0)
4
3
2
1
0
−5
−4
−3
−2
−1
0
1